

10. Rule-Based Classification vs Decision Tree

Your last question asks you to compare **Rule-based classification** and **Decision Tree classification** and plot their accuracy.

We'll use:

- **JRip** → Rule-based classification
- **J48** → Decision Tree

A. JRip

Steps

1. Open dataset in WEKA.
2. Click **Classify**.
3. Click **Choose**.
4. Select:

Rules → JRip

5. Select **10-fold cross-validation**.
6. Click **Start**.
7. Record accuracy.

B. J48

Steps1

1. Click **Choose**.
2. Select:

Trees → J48

3. Select **10-fold cross-validation**.
4. Click **Start**.
5. Record accuracy.

Accuracy Comparison Graph

You can use R for the graph.

Open **RStudio** → **New File** → **R Script**.

OUTPUT :

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier Choose **J48 -C 0.25 -M 2**

Test options
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) play
 Start Stop

Result list (right-click for options)
 10:17:06 - trees.J48

Classifier output

```

outlook = overcast: yes (2.0)
outlook = rainy: yes (4.0/1.0)

Number of Leaves : 3
Size of the tree : 4

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances      2          20 %
Incorrectly Classified Instances    8          80 %
Kappa statistic                    -0.6667
Mean absolute error                 0.75
Root mean squared error             0.7862
Relative absolute error             142.2414 %
Root relative squared error         147.0173 %
Total Number of Instances          10

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.200    0.867    0.200    0.200    0.200     -0.667   0.042    0.302    no
               0.000    0.667    0.000    0.000    0.000     -0.667   0.042    0.302    no

=== Confusion Matrix ===
 a b  <-- classified as
 2 4 | a = yes
 4 0 | b = no

```

Status OK Log

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier Choose **JRip -F 3 -N 2.0 -O 2 -S 1**

Test options
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) play
 Start Stop

Result list (right-click for options)

Classifier output

```

=====
(outlook = sunny) => play=no (4.0/1.0)
=> play=yes (6.0/1.0)

Number of Rules : 2

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances      3          30 %
Incorrectly Classified Instances    7          70 %
Kappa statistic                    -0.5217
Mean absolute error                 0.66
Root mean squared error             0.7385
Relative absolute error             125.1724 %
Root relative squared error         138.0974 %
Total Number of Instances          10

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Clas
Weighted Avg.   0.500    1.000    0.429    0.500    0.462     -0.535   0.375   0.675   yes
               0.000    0.500    0.000    0.000    0.000     -0.535   0.375   0.475   no

=== Confusion Matrix ===
 a b  <-- classified as
 3 3 | a = yes
 4 0 | b = no

```

Status OK Log

